

Problem statement 2: Person and Student

- Base $\rightarrow$ Person $\rightarrow$ store name
- derived $\rightarrow$ student $\rightarrow$ store mark

Tasks : Same. But using Dynamically

Problem statement 3: Shape and Rectangle

- Base $\rightarrow$ ~~Area~~ shape $\rightarrow$ Area() or "finding area" by length + breadth and calculate
- derived $\rightarrow$ rectangle $\rightarrow$

Example

```
Base* bptr;  
Derived d;  
bptr = d;  
bptr → show(); // work  
// bptr → display(); // error
```

Why use pointer to derived class.

- when you want full access to derived object
- when working with specific child class feature
- Useful in dynamic memory allocation

Dynamic Memory Example.

```
Derived *ptr = new Derived();
```

```
ptr → show();
```

```
ptr → display();
```

```
delete ptr;
```

Problem statement 1: Animal and Dog

design a program using inheritance.

- Base → Animal → Eat()
- derived → Dog → Bark()

Task: Create a obj of Dog and use pointer to derived class to call both eat() and bark()


```

public:
    void show() {
        cout << "Base class function" << endl;
    }
};

```

```

class Derived : public Base {
public:
    void display() {
        cout << "Derived class function" << endl;
    }
};

```

```

int main() {
    Derived d;
    Derived *ptr = &d; // pointer to derived class.

    ptr -> show();
    ptr -> display();

    return 0;
}

```

Pointer type derived can access

- Base class member (inherit)
- Derived class member

Important Comparison

• Pointer to derived class can access both
Base and derived class

```

    this -> x = x;
    return *this;
}

```

Method chaining (future)

- Problem statement 1: student marks
- Problem statement 2: Rectangle dimension.
- Problem statement 3: Employee detail;

Day 42 (Pointers to derived class)

A pointer to derived class is a pointer that stores the address of an object of a derived class.

Basic Example

```

#include <iostream>
using namespace std;

```